

CALCULUS (II)

WEEK (III)

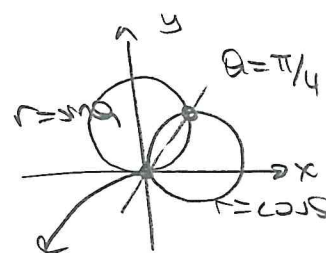
INTERSECTIONS OF POLAR CURVES

→ Because the polar coordinates of points are not unique, finding the intersection points of two polar curves can be more complicated than the similar problem for Cartesian graphs.

→ Of course, the polar curves $r = f(\theta)$ and $r = g(\theta)$ will intersect at any point $[r_0, \theta_0]$ for which

$$f(\theta_0) = g(\theta_0) \text{ and } r_0 = f(\theta_0)$$

but there may be other intersections as well



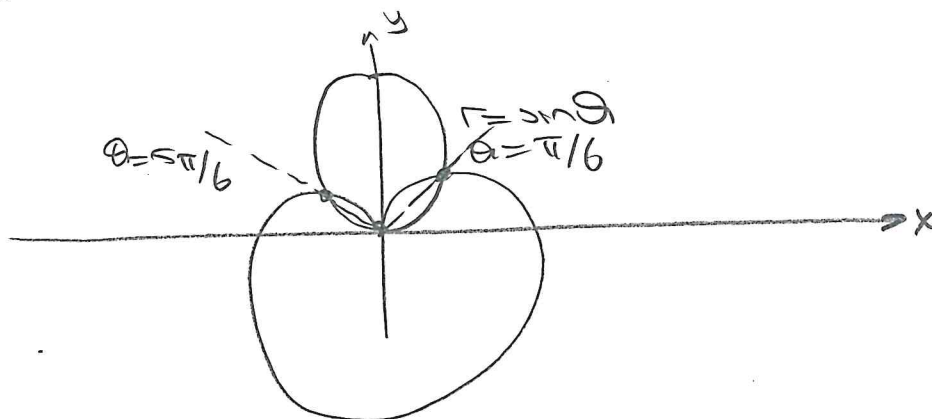
!! This point $(0,0)$ can't be obtained from $\cos \theta = \sin \theta$

Ex 7 Find the intersections of the curves $r = \sin \theta$ and $r = 1 - \sin \theta$. Since both functions of θ are periodic with period 2π , we need only look for solutions satisfying $0 \leq \theta \leq 2\pi$

$$\sin \theta = 1 - \sin \theta$$

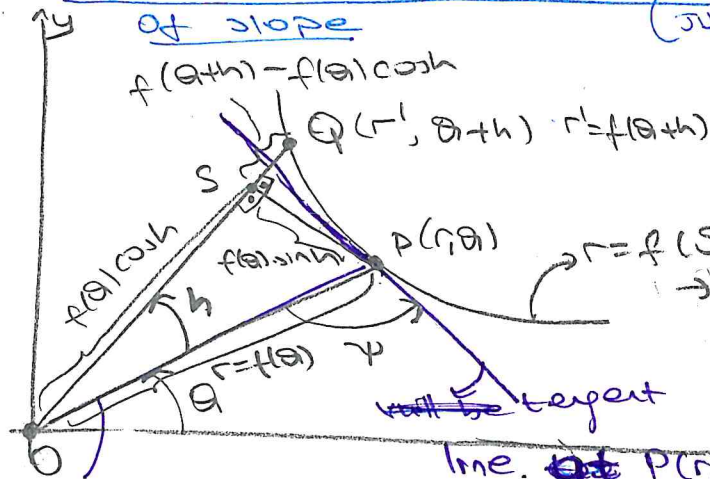
$$\Rightarrow \sin \theta = 1/2$$

$$\theta = \pi/6 \text{ or } \theta = 5\pi/6$$



SLOPES, AREAS AND ARC LENGTHS FOR POLAR CURVES

Geometric Interpretation (NOT IMPORTANT FOR STUDENTS)
↳ of slope (JUST PAGE 2)



Consider the curve $r = f(\theta)$

$$PS = f(\theta) \cdot \sin h$$

$$SQ = OQ - OS = f(\theta + h) - f(\theta) \cos$$

→ If the tangent line to $r = f(\theta)$ makes angle γ with

radial line O.P.

Then, The angle ψ is the
X limit of $\sin \psi$ as $h \rightarrow 0$ and

Line. ~~of~~ $P(r, \theta)$

we have $\tan \gamma = \lim_{h \rightarrow 0} \frac{PS}{SQ} = \lim_{h \rightarrow 0} \frac{f(1)\sinh}{f(1)\cosh - f(1)\sinh} \quad (0/0)$

Apply L'H
 $\lim_{h \rightarrow 0} \frac{f(\theta) \cosh h}{f'(\theta + h) + f(\theta) \sinh h} = \frac{f(\theta)}{f'(\theta)} = \frac{r}{dr/d\theta}$

Repeat direction for a polar curve

At any point P other than the origin on the polar curve $r = f(\theta)$, the angle ψ between the radial line from the origin to P and the tangent to the

Curve is given by

$$\tan \psi = \frac{f(\theta)}{f'(\theta)}$$

→ Assume that the equation is in the form $r = f(\theta)$.

Recall the conversion equation for going from polar coordinates to cartesian coordinates

$$x = r \cos \theta \quad y = r \sin \theta$$

→ Since, we assume that $r = f(\theta)$ we have

$$x = f(\theta) \cos \theta, \quad y = f(\theta) \sin \theta.$$

→ So,

$$\frac{dx}{d\theta} = f'(\theta) \cos \theta - f(\theta) \sin \theta = \frac{dr}{d\theta} \cos \theta - r \sin \theta$$

$$\frac{dy}{d\theta} = f'(\theta) \sin \theta + f(\theta) \cos \theta = \frac{dr}{d\theta} \sin \theta + r \cos \theta$$

→ Therefore, we have

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} \quad \frac{dx}{d\theta} \neq 0$$

Horizontal
Tangents

$$\frac{dy}{d\theta} = 0$$

provided that $\frac{dx}{d\theta} \neq 0$

Vertical
Tangents

$$\frac{dx}{d\theta} = 0$$

provided that $\frac{dy}{d\theta} \neq 0$

NOTE If we are looking for tangent lines at the pole,
then $r = 0$ and we have

$$\frac{dy}{dx} = \tan \theta \quad \text{if} \quad \frac{dr}{d\theta} \neq 0$$

Ex 7 Find the points on the cardioid $r = 1 + \cos \theta$ where the tangent lines are vertical or horizontal
 $r = 1 + \cos \theta$ using $y = r \sin \theta$ $x = r \cos \theta$, we obtain
 $y = (1 + \cos \theta) \sin \theta$ $x = (1 + \cos \theta) \cos \theta$

For horizontal tangents, we have

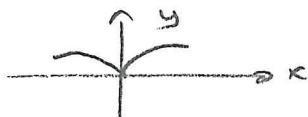
$$\begin{aligned} 0 = \frac{dy}{d\theta} &= -\sin^2 \theta + \cos^2 \theta + \cos \theta \\ &= 2\cos^2 \theta + \cos \theta - 1 \\ &= (2\cos \theta - 1)(\cos \theta + 1) \end{aligned}$$

The solutions are $\cos \theta = 1/2$ and $\cos \theta = -1$, that is, $\theta = \pm \pi/3$ and $\theta = \pi$. There are horizontal tangents at $[\frac{3}{2}, \pm \pi/3]$.

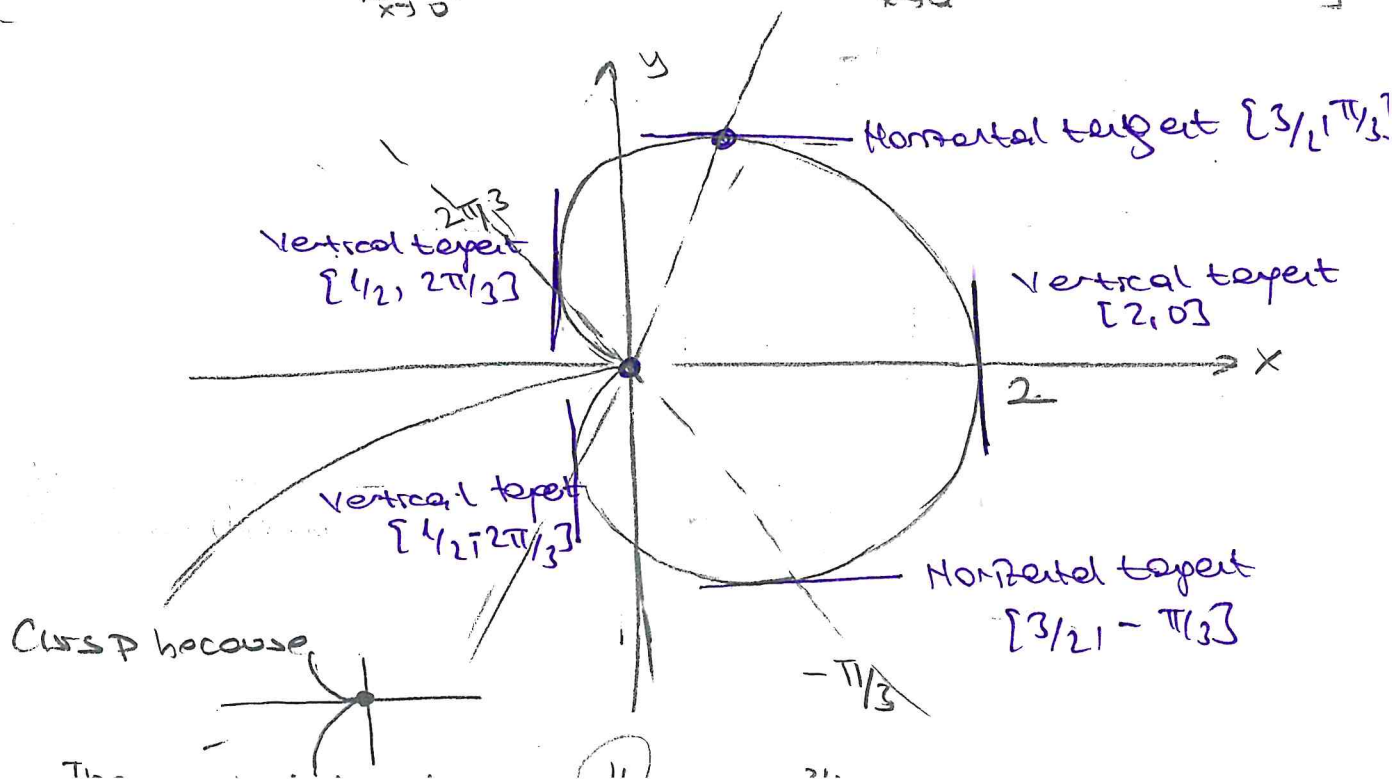
At $\theta = \pi$, we have $r = 0$. The curve doesn't have a tangent line at the origin (it has a cusp)

Cusp derivative is zero at this point but directional derivatives has different signs in different sides

$y = x^{2/3}$
 $(a, f(a))$
 $a = 0$ for $y = x^{2/3}$
 $y' = \frac{2}{3}x^{-1/3}$ $\lim_{x \rightarrow 0^+} y' = \infty$
 $\lim_{x \rightarrow 0^-} y' = -\infty$



① $\lim_{x \rightarrow a^+} f'(x) = \infty$
 $\lim_{x \rightarrow a^-} f'(x) = -\infty$
 OR
 ② $\lim_{x \rightarrow a^+} f'(x) = -\infty$
 $\lim_{x \rightarrow a^-} f'(x) = \infty$



For vertical tangents we will have

$$\frac{dx}{d\theta} = -\sin\theta - 2\cos\theta \sin\theta = -\sin\theta (1 + 2\cos\theta)$$

The solutions are $\sin\theta = 0$ and $\cos\theta = -1/2$, that is

$\theta = 0, \pi, \pm 2\pi/3$. There are vertical tangents at $[2, 0]$ at $[1/2, \pm 2\pi/3]$

Ex 7 (a) For the cardioid $r = 1 + \sin\theta$, find the slope of the tangent line when $\theta = \pi/3$

(b) Find the points on the cardioid where the tangent line is horizontal and vertical

$$r = 1 + \sin\theta$$

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin\theta + r \cos\theta}{\frac{dr}{d\theta} \cos\theta - r \sin\theta}$$

$$y = r \sin\theta$$

$$x = r \cos\theta$$

$$\frac{dr}{d\theta} = \cos\theta$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\cos\theta \cdot \sin\theta + (1 + \sin\theta) \cos\theta}{\cos\theta \cos\theta - (1 + \sin\theta) \sin\theta} \\ &= \frac{\cos\theta + 2\cos\theta \sin\theta}{\cos^2\theta - \sin^2\theta - \sin\theta} = \frac{\cos\theta + \sin 2\theta}{\cos 2\theta - \sin\theta} \end{aligned}$$

$$\textcircled{a} \left. \frac{dy}{dx} \right|_{\theta=\pi/3} = \frac{\cos \pi/3 + \sin 2\pi/3}{\cos 2\pi/3 - \sin \pi/3} = \frac{1/2 + \sqrt{3}/2}{-1/2 - \sqrt{3}} = -1 \quad y - \frac{3+\sqrt{3}}{4} = -1 \left(x - \frac{2+\sqrt{3}}{4} \right)$$

Equation of tangent

$$\begin{aligned} x &= r \cos\theta = (1 + \sin\theta) \cos\theta = (1 + \sqrt{3}/2) \cdot 1/2 = (2 + \sqrt{3})/4 \\ y &= r \sin\theta = (1 + \sin\theta) \sin\theta = (1 + \sqrt{3}/2) \sqrt{3}/2 = \frac{3 + \sqrt{3}}{4} \end{aligned}$$

$$\textcircled{b} \text{ Horizontal } \frac{dy}{d\theta} = \frac{dr}{d\theta} \sin\theta + r \cos\theta = 0$$

$$= \cos\theta (\sin\theta + \frac{1 + \sin\theta}{r}) = 0$$

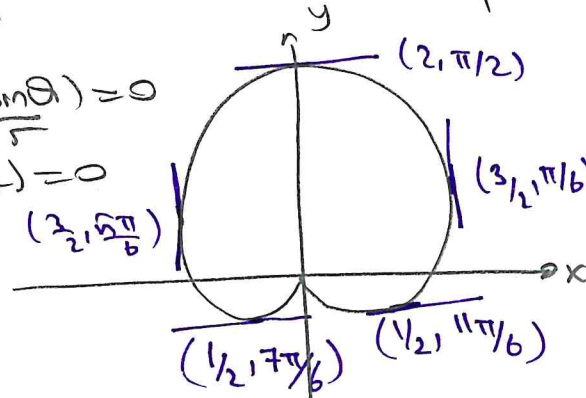
$$= \cos\theta (2\sin\theta + 1) = 0$$

$$\cos\theta = 0$$

$$\theta = \pi/2, \theta = 3\pi/2$$

$$1 + 2\cos\theta = 0$$

$$\theta = 7\pi/6, \theta = 11\pi/6$$



$(2, \pi/2)$ $(1/2, 7\pi/6)$ $(1/2, 11\pi/6)$ $(3/2, \pi/6)$ $\rightarrow$ Horizontal tangent lines

Vertical Tangent lines

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{dr}{d\theta} \cos\theta - r \sin\theta = 0 \\ &= \cos^2\theta - (1+\sin\theta)\sin\theta \\ &= (1-\sin^2\theta) - (1+\sin\theta)\sin\theta \\ &= (1+\sin\theta)(1-2\sin\theta)\end{aligned}$$

$$1+\sin\theta=0 \Rightarrow \theta = 3\pi/2$$

$$(3/2, \pi/6), (3/2, 5\pi/6)$$

$$1-2\sin\theta=0 \Rightarrow \theta = \pi/6, \theta = 5\pi/6$$

$$(0, 3\pi/2) \text{ cusp}$$

→ Be careful about $\theta = 3\pi/2$. When $\theta = 3\pi/2$, both $\frac{dx}{d\theta}$ and

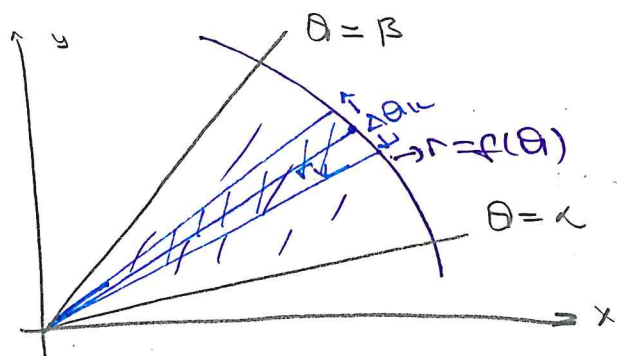
$\frac{dy}{d\theta}$ is zero. Now let's look for limit

$$\begin{aligned}\lim_{\theta \rightarrow (3\pi/2)^-} \frac{dy}{dx} &= \left(\lim_{\theta \rightarrow 3\pi/2} \frac{1+2\sin\theta}{1-\sin\theta} \right) \left(\lim_{\theta \rightarrow 3\pi/2} \frac{\cos\theta}{1+\sin\theta} \right) \\ &= -\frac{1}{3} \lim_{\theta \rightarrow 3\pi/2} \frac{-\sin\theta}{\cos\theta} = \infty\end{aligned}$$

$$\lim_{\theta \rightarrow (3\pi/2)^+} \frac{dy}{dx} = -\infty$$

$$\theta = 3\pi/2 \text{ is a cusp}$$

AREAS AND ARC LENGTHS FOR POLAR CURVES



The area of the shaded region

is

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta \quad (*)$$

Why we obtain this formula $(*)$?

Roughly speaking we divide this region into fan-shaped regions (blue part in figure). The areas of this blue parts

approximately given by

Area of a circle with radius r

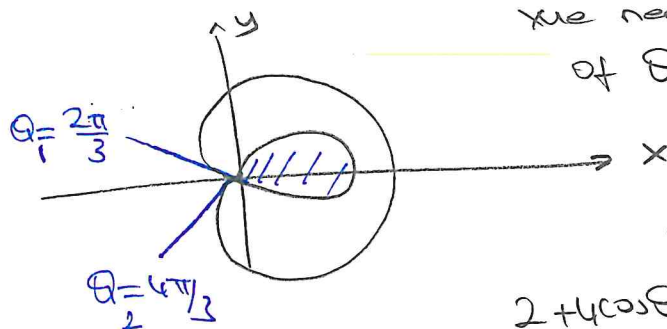
$$A_k = \frac{\Delta\theta_k}{2\pi} (\pi r_k^2) = \frac{1}{2} r_k^2 \Delta\theta_k = \frac{1}{2} (f(\theta_k))^2 \Delta\theta_k$$

Then the area of the entire region is approximately

$$\sum_{k=1}^n A_k = \sum_{k=1}^n \frac{1}{2} (f(\theta_k))^2 \Delta\theta_k$$

$$A = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \frac{1}{2} (f(\theta_k))^2 \Delta\theta_k = \int_{\alpha}^{\beta} \frac{1}{2} (f(\theta))^2 d\theta$$

Ex 7 Determine the area of the inner loop of $r = 2 + 4\cos\theta$.
(See page (10) on week 11 lectures).



We need to know the values of θ where the curve goes through the origin we get this values by setting the equation equal to zero

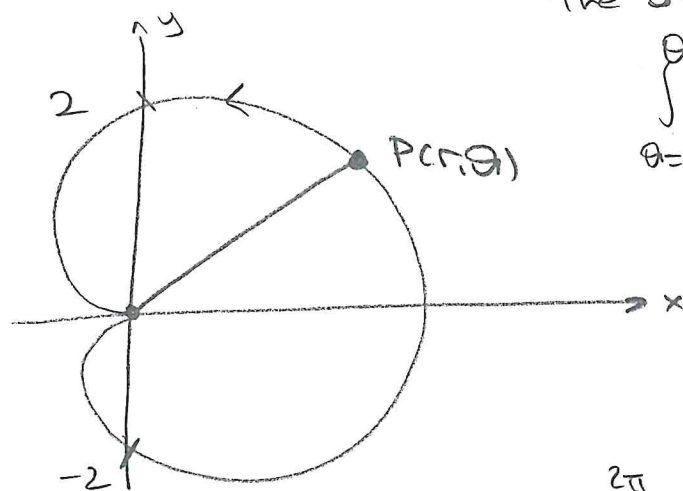
$$2 + 4\cos\theta = 0 \quad \cos\theta = -1/2 \quad \theta_1 = \frac{2\pi}{3}, \theta_2 = \frac{4\pi}{3}$$

These points define where the inner loop starts and ends. So,

$$A = \int_{2\pi/3}^{4\pi/3} \frac{1}{2} (2 + 4\cos\theta)^2 d\theta = \int_{2\pi/3}^{4\pi/3} \frac{1}{2} (4 + 16\cos\theta + 16\cos^2\theta) d\theta = 4\pi - 6\sqrt{3}$$

(7)

Ex 7 Find the area of the region in the plane enclosed by the cardioid $r = 2(1 + \cos \theta)$



The area is

$$\int_{\theta=0}^{\theta=2\pi} \frac{1}{2} r^2 d\theta = \int_0^{2\pi} \frac{1}{2} 4 (1 + \cos \theta)^2 d\theta$$

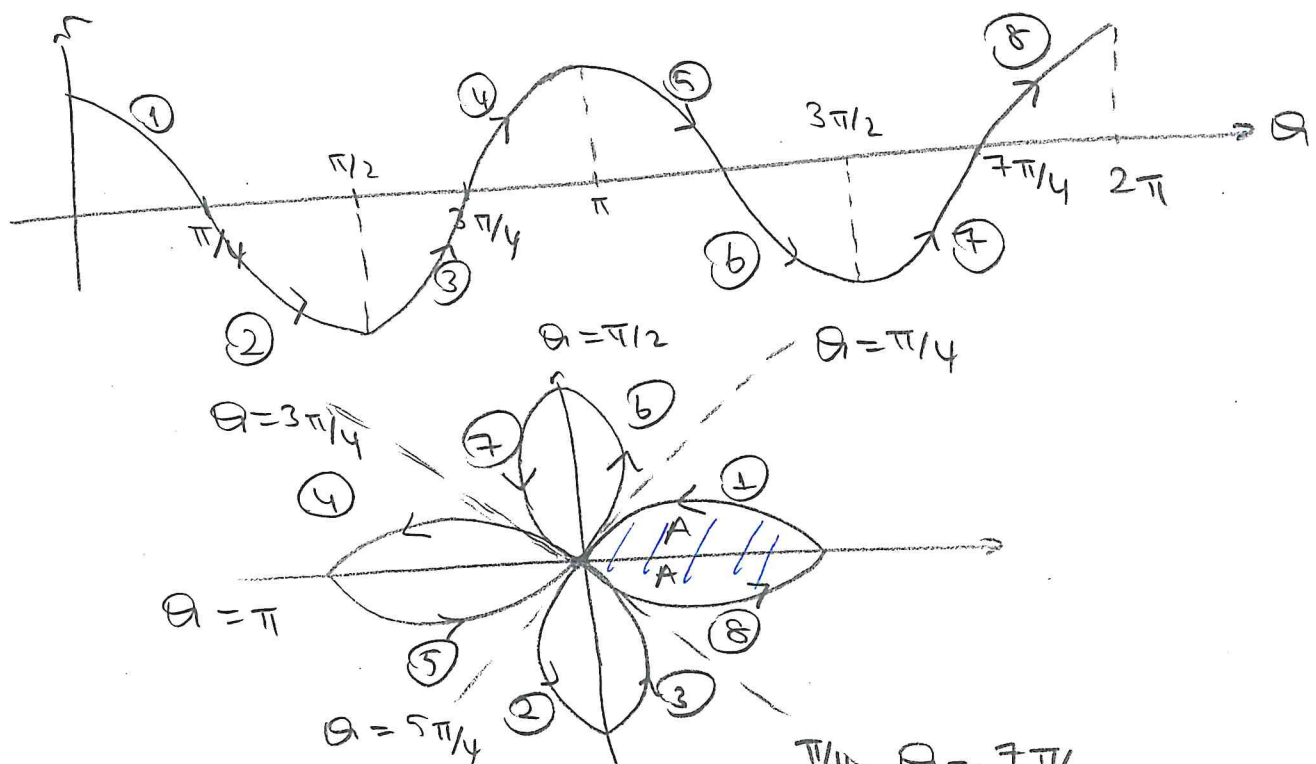
$$= \int_0^{2\pi} 2 (1 + 2\cos \theta + \cos^2 \theta) d\theta$$

$$= \int_0^{2\pi} \left(2 + 4\cos \theta + 2 \cdot \frac{1 + \cos 2\theta}{2} \right) d\theta$$

$$= \int_0^{2\pi} (3 + 4\cos \theta + \cos 2\theta) d\theta$$

$$= \left[3\theta + 4\sin \theta + \frac{\sin 2\theta}{2} \right]_{\theta=0}^{\theta=2\pi} = 6\pi - 0 = 6\pi$$

Ex Find the area enclosed by one loop of the four-leaved rose $r = \cos 2\theta$



Area of one loop = $2A = 2 \cdot \frac{1}{2} \int_{\theta=0}^{\theta=\pi/4} \cos^2 2\theta d\theta$

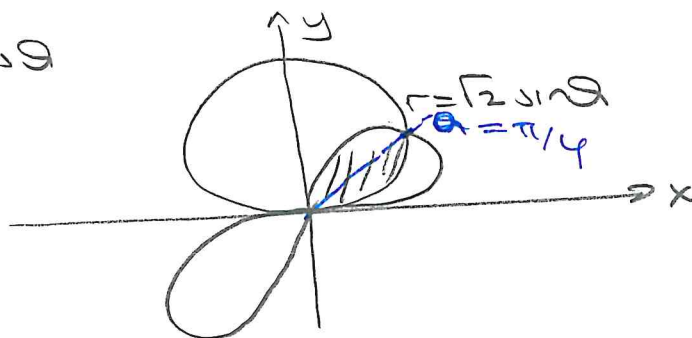
$$= \int_0^{\pi/4} \frac{1}{2} (1 + \cos 4\theta) d\theta = \left[\frac{\theta}{2} + \frac{1}{4} \sin 4\theta \right]_0^{\pi/4} = \frac{\pi}{8}$$

Ex 7 Find the area of the region that lies inside the circle $r = \sqrt{2} \sin \theta$ and inside the lemniscate $r^2 = \sin 2\theta$

$$2 \sin^2 \theta = \sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin \theta = \cos \theta$$

$$\theta = \pi/4$$



The required area is

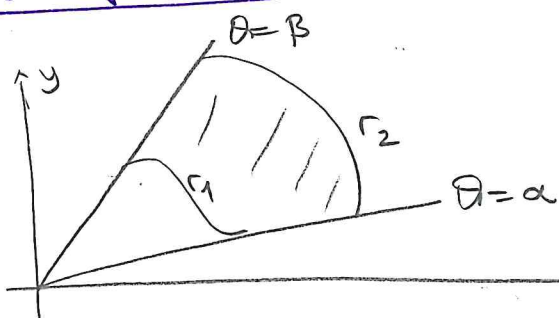
$$A = \frac{1}{2} \int_0^{\pi/4} 2 \sin^2 \theta d\theta + \frac{1}{2} \int_{\pi/4}^{\pi/2} \sin 2\theta d\theta$$

$$= \int_0^{\pi/4} \frac{1 - \cos 2\theta}{2} d\theta - \frac{1}{4} \cos 2\theta \Big|_{\pi/4}^{\pi/2}$$

$$= \frac{\pi}{8} - \frac{1}{4} \sin 2\theta \Big|_0^{\pi/4} + \frac{1}{4} = \frac{\pi}{8} - \frac{1}{4} + \frac{1}{4} = \frac{\pi}{8} \text{ square units}$$

AREA BETWEEN CURVES

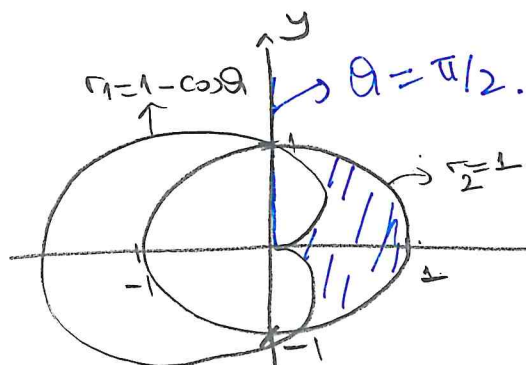
Area of the Region $0 \leq r_1(\theta) \leq r \leq r_2(\theta) \quad \alpha \leq \theta \leq \beta$



$$A = \int_{\alpha}^{\beta} \frac{1}{2} r_2^2 d\theta - \int_{\alpha}^{\beta} \frac{1}{2} r_1^2 d\theta$$

$$A = \int_{\alpha}^{\beta} \frac{1}{2} (r_2^2 - r_1^2) d\theta$$

Ex 7 Find the area of the region that lies inside the circle $r = 1$ and outside the cardioid $r = 1 - \cos \theta$



$$1 = r = 1 - \cos \theta$$

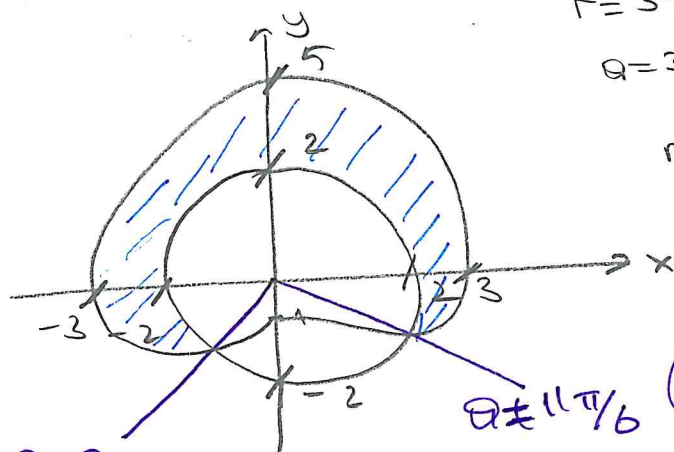
$$\cos \theta = 0 \Rightarrow \theta = \pi/2 + k\pi \quad k \in \mathbb{Z}$$

$$A = \int_{-\pi/2}^{\pi/2} \frac{1}{2} (r_2^2 - r_1^2) d\theta$$

$$= 2 \int_0^{\pi/2} \frac{1}{2} (r_2^2 - r_1^2) d\theta$$

$$= \int_0^{\pi/2} (2 \cos \theta - \cos^3 \theta) d\theta = \int_0^{\pi/2} \left(2 \cos \theta - \frac{1 + \cos 2\theta}{2} \right) d\theta = \left[2 \sin \theta - \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\pi/2}$$

Ex 7 Determine the area that lies inside $r=3+2\sin\theta$ and outside $r=2$



$$r = 3 + 2\sin\theta$$

$$a = 3 > b = 2$$

Limaçon without an inner loop

$r = 2$ circle with $M(0,0)$ radius 2

$$\theta \neq 11\pi/6 \quad (\theta = -\pi/6!!)$$

$$\theta = 7\pi/6$$

→ we first determine the values of θ for which the two curves intersect.

$$3 + 2\sin\theta = 2 \Rightarrow \sin\theta = -1/2$$

$$\theta = 7\pi/6 \quad \text{and} \quad \theta = 11\pi/6$$

→ Another representation of $\frac{11\pi}{6}$ is $-\pi/6$. In order to use the formula above the area must be enclosed as we increase from smaller to larger angle. So if we use $7\pi/6$ to $11\pi/6$ we will not enclose the shaded area, instead we will enclose the bottom shaded area, instead we will enclose the bottom most of three regions. However, if we use $-\pi/6$ to $\frac{7\pi}{6}$ we will enclose the shaded area, so,

$$A = \int_{-\pi/6}^{7\pi/6} \frac{1}{2} \left((3 + 2\sin\theta)^2 - 2^2 \right) d\theta = \frac{11\sqrt{3}}{2} + \frac{14\pi}{3}$$

LENGTH OF A POLAR CURVE

→ we can obtain a polar coordinate formula for the length of a curve $r = f(\theta)$ $\alpha \leq \theta \leq \beta$ by parametrizing the curve as

$$x = r \cos \theta = f(\theta) \cos \theta \quad \alpha \leq \theta \leq \beta$$

$$y = r \sin \theta = f(\theta) \sin \theta$$

→ The parametric length formula then gives the length as

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

This equation becomes

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Derivation

$$x = r \cos \theta = f(\theta) \cos \theta$$

$$y = r \sin \theta = f(\theta) \sin \theta$$

$$\frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta - r \sin \theta$$

$$\frac{dy}{d\theta} = \frac{dr}{d\theta} \sin \theta + r \cos \theta$$

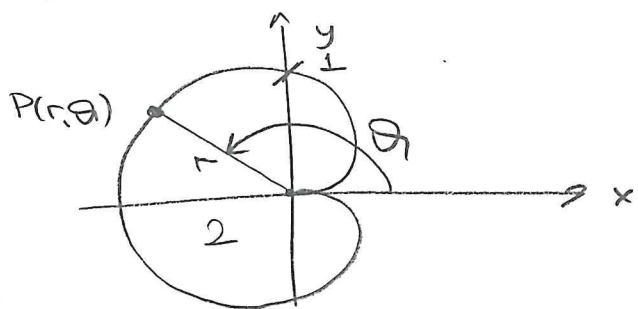
$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= \left(\frac{dr}{d\theta} \cos \theta - r \sin \theta\right)^2 + \left(\frac{dr}{d\theta} \sin \theta + r \cos \theta\right)^2 \\ &= \left(\frac{dr}{d\theta}\right)^2 \cos^2 \theta - 2r \frac{dr}{d\theta} \cos \theta \sin \theta + r^2 \sin^2 \theta \\ &\quad + \left(\frac{dr}{d\theta}\right)^2 \sin^2 \theta + 2r \cos \theta \sin \theta \frac{dr}{d\theta} + r^2 \cos^2 \theta \end{aligned}$$

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \left(\frac{dr}{d\theta}\right)^2 + r^2$$

Arc Length in Cartesian

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Ex 7 Find the length of the cardioid $r = 1 - \cos \theta$



$$r = 1 - \cos \theta \quad \frac{dr}{d\theta} = \sin \theta$$

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= (1 - \cos \theta)^2 + (\sin \theta)^2 \\ &= 1 - 2\cos \theta + \underbrace{\cos^2 \theta + \sin^2 \theta}_1 \\ &= 2 - 2\cos \theta \end{aligned}$$

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{2 - 2\cos \theta} d\theta = \int_0^{2\pi} \sqrt{4 \sin^2 \frac{\theta}{2}} d\theta$$

$1 - \cos \theta = 2 \sin^2(\theta/2)$

$$= \int_0^{2\pi} 2 \left| \sin \frac{\theta}{2} \right| d\theta = \int_0^{2\pi} 2 \sin \frac{\theta}{2} d\theta = \left[-4 \cos \frac{\theta}{2} \right]_0^{2\pi} = 8$$

$$\sin(\theta/2) \geq 0 \text{ for } 0 \leq \theta \leq 2\pi$$

Ex Determine the length of $r = \theta$ for $0 \leq \theta \leq 1$

$$\frac{dr}{d\theta} = 1 \quad L = \int_0^1 \sqrt{\theta^2 + 1} d\theta = \int_0^{\pi/4} \sec^3 x dx$$

Change of variables

$$\theta = \tan x \quad d\theta = \sec^2 x dx$$

$$\theta = 0 \Rightarrow x = 0$$

$$\theta = 1 \Rightarrow x = \pi/4$$

$$\sqrt{\theta^2 + 1} = \sqrt{\tan^2 x + 1} = |\sec x| = \sec x$$

$$\int_0^{\pi/4} \sec^3 x dx$$

Integration by part

$$u = \sec x \quad \sec^2 x dx = dv$$

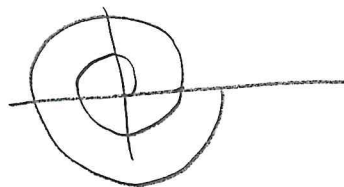
$$du = \sec x \tan x dx \quad \tan x = v$$

$$\int u dv = uv - \int v du$$

NOTE The polar equation

$r = \theta$ is the equation of a spiral

$$r = \theta \text{ for } 0 \leq \theta \leq 1$$



$$L = \int_0^{\pi/4} \sec^3 x \, dx = \sec x \tan x \Big|_0^{\pi/4} - \int_0^{\pi/4} \sec x \underbrace{\tan^2 x}_{\sec^2 x - 1} \, dx$$

$\sec x = u$
 $\sec^2 x \, dx = du$
 $\sec x \tan x = du$
 $\tan x = v$

$$= \underbrace{\sec x \tan x \Big|_0^{\pi/4}}_{\frac{2}{\sqrt{2}}} - \int_0^{\pi/4} \sec^3 x \, dx + \int_0^{\pi/4} \sec x \, dx \underbrace{\frac{\sec x + \tan x}{\sec x + \tan x}}_{(\sec x)^1}$$

$$\left(\ln(\sec x + \tan x) \right)' = \frac{\underbrace{(\sec x)'}_{\sec x \tan x} + \underbrace{(\tan x)'}_{\sec^2 x}}{\sec x + \tan x}$$

$$2 \underbrace{\int_0^{\pi/4} \sec^3 x \, dx}_L = \frac{2}{\sqrt{2}} + \ln |\sec x + \tan x| \Big|_0^{\pi/4}$$

$$L = \frac{1}{\sqrt{2}} + \frac{1}{2} \ln \left| \frac{2}{\sqrt{2}} + 1 \right|$$

SURFACE AREA WITH POLAR COORDINATES

→ The surface area of the region found by rotating

$$r = f(\theta) \quad \alpha \leq \theta \leq \beta \quad \text{is}$$

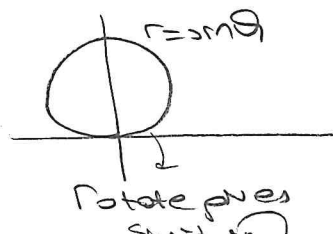
Rotated x-axis (Polar axis)

$$S = \int 2\pi \underbrace{r \sin \theta}_{y} \, ds \quad \left\{ \begin{array}{l} ds = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \, d\theta \\ r = f(\theta) \end{array} \right.$$

Rotated y-axis ($\theta = \pi/2$)

$$S = \int 2\pi \underbrace{r \cos \theta}_{x} \, ds$$

Ex Determine the surface area formed by the rotation of $r = \sin \theta$ about polar axis



$$r^2 = \sin^2 \theta$$

$$\frac{dr}{d\theta} = \cos \theta$$

$$S = 2\pi \int_0^{\pi} \underbrace{\sin \theta}_{y} \underbrace{\sin \theta}_{r} \sqrt{\sin^2 \theta + \cos^2 \theta} \, d\theta$$

Circle of $r = \sin \theta$

$$= 2\pi \int_0^{\pi} \sin^2 \theta \, d\theta = \pi \int_0^{\pi} (1 - \cos 2\theta) \, d\theta = \frac{\pi}{2} (\theta - \sin 2\theta) \Big|_0^{\pi} = \frac{\pi^2}{2}$$

